

MID REVIEW REPORT

Customer Retention and Sales Analysis using Online Retail Data

Role: EDA & Visualization

Introduction

The objective of this project is to analyze customer purchasing behavior, product performance, revenue trends, and country-wise sales performance using Exploratory Data Analysis (EDA). The project aims to identify business trends and generate insights that support data-driven decision making.

Week 1 Progress Report – Exploratory Data Analysis (EDA)

Work Completed

- Dataset Understanding (EDA)
- Transaction Analysis (EDA)
- Customer Analysis (EDA)
- Country Analysis (EDA)
- Initial Business Insights

Activities Performed:

1. Dataset Understanding

- Loaded and explored the Online Retail dataset.
- Examined dataset structure, columns, and data types.
- Reviewed dataset information and overall data quality.

2. Transaction Analysis

- Calculated the number of unique invoices/orders.
- Analyzed transaction activity across countries.
- Identified top countries by transaction count.

3. Customer Analysis

- Calculated the number of unique customers.
- Analyzed customer distribution across countries.
- Identified countries with the highest number of customers.

4. Country Analysis

- Compared country-wise transaction and customer activity.
- Identified major countries contributing to business performance.
- Created visualizations for country-level analysis.

5. Initial EDA Findings

- Studied customer purchasing behavior.
- Documented observations and trends from the analysis.

Key Insights

- Customer activity is concentrated in a few countries.
- Transaction volume varies significantly across countries.
- Customer distribution patterns help identify key markets.
- EDA provided valuable insights into customer behavior.

Week 2 Progress Report – EDA & Visualization

Work Completed

- Top Products Analysis (EDA + Visualization)
- Revenue Analysis (EDA + Visualization)
- Top Countries Revenue Analysis
- Monthly Revenue Trend Analysis
- Top Customers Analysis
- Business Insights and Documentation

Activities Performed:

1. Product Analysis

- Identified top 10 products by quantity sold.
- Identified top 10 products by revenue generated.
- Created product performance visualizations.

2. Revenue Analysis

- Calculated Revenue using $\text{Quantity} \times \text{Unit Price}$.
- Analyzed top revenue-generating products and countries.
- Created revenue-based visualizations.

3. Monthly Revenue Trend Analysis

- Converted InvoiceDate into datetime format.
- Performed month-wise revenue analysis.

- Created Monthly Revenue Trend chart.

4. Top Customers Analysis

- Identified top 10 customers by revenue contribution.
- Analyzed customer value and purchasing behavior.
- Created customer comparison visualizations.

5. Business Insights & Documentation

- Added observations below charts.
- Prepared Executive Summary.
- Prepared Business Recommendations and Conclusion.

Key Insights

- United Kingdom generated the highest transactions and revenue.
- A small number of products contributed significantly to total sales.
- Top customers generated a major share of revenue.
- Monthly revenue trends highlighted business performance patterns.

Deliverables Completed

- Dataset Understanding (EDA)
- Transaction Analysis
- Customer Analysis
- Country Analysis
- Top Products by Quantity Sold
- Top Products by Revenue
- Top Countries by Revenue
- Monthly Revenue Trend Analysis
- Top Customers Analysis
- Business Insights
- Executive Summary
- Business Recommendations
- Conclusion

Current Project Status

All assigned EDA and Visualization tasks have been completed successfully. The project includes customer analysis, transaction analysis, product analysis, country analysis, revenue analysis, visualizations, business insights, recommendations, and conclusion.